Toward an Averaged GEqOE Formulation for Autonomous Zonal Dynamics

J_2 Closure, General Zonal Structure, and a Path to Fourier-in-K Thrust Averaging

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Abstract

This note specializes the mean / averaged GEqOE question to the autonomous conservative case generated by the Earth zonal harmonics. The objective is to identify a tractable research path toward a fast analytical or semi-analytical model that could later support maneuver-detection metrics.

Three results are recorded. First, for a static zonal potential, the current GEqOE equations already admit an exact first-order orbit-averaging operator in the propagated generalized eccentric longitude K. Second, for J_2 alone, the first-order secular closure is especially simple: ν, the generalized eccentricity magnitude $g = \sqrt{p_1^2 + p_2^2}$, and the inclination magnitude $Q = \sqrt{q_1^2 + q_2^2}$ are constants, while the pairs (p_1, p_2) and (q_1, q_2) undergo uniform rotations with explicit secular rates. Third, the same K-based framework provides a clean insertion point for a future Fourier-in-K thrust averaging theory, provided the first step is taken around the autonomous J_2 or zonal baseline.

1 Scope

The present note focuses on the GEqOE state propagated in the current branch:

$$\boldsymbol{y} = (\nu, p_1, p_2, K, q_1, q_2). \tag{1}$$

The working assumptions are:

1. the conservative perturbation is autonomous and axisymmetric about the inertial z axis;
2. the disturbing potential is either the pure J_2 term or a static zonal expansion;
3. averaging is performed with respect to the propagated fast angle K;
4. only first-order secular dynamics are sought at this stage.

This restriction is deliberate. It keeps the averaged problem autonomous and avoids the additional complications introduced by explicit time dependence (third-body ephemerides, rotating-frame geopotential tides, etc.).

2 GEqOE Preliminaries

Define the auxiliary quantities

$$g^2 = p_1^2 + p_2^2, \qquad \beta = \sqrt{1 - g^2}, \qquad \alpha = \frac{1}{1+\beta}, \tag{2}$$

$$a = \left(\frac{\mu}{\nu^2}\right)^{1/3}, \tag{3}$$

$$Q^2 = q_1^2 + q_2^2, \qquad \gamma = 1 + Q^2, \qquad \delta = 1 - Q^2. \tag{4}$$

From the equinoctial orientation variables one has

$$\cos i = \frac{\delta}{\gamma}, \qquad \tan\frac{i}{2} = Q. \tag{5}$$

The propagated generalized eccentric longitude K gives

$$r = a\left(1 - p_1 \sin K - p_2 \cos K\right), \tag{6}$$

$$X = a\left(\alpha p_1 p_2 \sin K + (1 - \alpha p_1^2)\cos K - p_2\right), \tag{7}$$

$$Y = a\left(\alpha p_1 p_2 \cos K + (1 - \alpha p_2^2)\sin K - p_1\right), \tag{8}$$

and the normalized inertial z coordinate is

$$\hat{z} = \frac{2(Y q_2 - X q_1)}{\gamma r}. \tag{9}$$

The generalized angular momentum proxy used throughout the GEqOE paper and the current implementation is

$$c = \left(\frac{\mu^2}{\nu}\right)^{1/3} \beta. \tag{10}$$

Under purely conservative autonomous dynamics, the generalized energy is a first integral and therefore

$$\dot{\nu} = 0. \tag{11}$$

3 Autonomous Zonal GEqOE Dynamics

For a zonal potential of the form

$$U(r, \hat{z}) = \sum_{n \geq 2} U_n(r, \hat{z}) = \sum_{n \geq 2} \frac{C_n}{r^{n+1}} P_n(\hat{z}), \qquad C_n = \mu J_n R_e^n, \tag{12}$$

the current branch uses the exact conservative GEqOE system

$$\dot{\nu} = 0, \tag{13}$$

$$\dot{p}_1 = p_2(d - w_h) + \frac{1}{c}\left(\frac{X}{a} + 2p_2\right)\mathcal{E}_Z, \tag{14}$$

$$\dot{p}_2 = p_1(w_h - d) - \frac{1}{c}\left(\frac{Y}{a} + 2p_1\right)\mathcal{E}_Z, \tag{15}$$

$$\dot{K} = \frac{w}{r} + d - w_h + \frac{1}{c}\left(1 + \alpha\left(1 - \frac{r}{a}\right)\right)\mathcal{E}_Z, \tag{16}$$

$$\dot{q}_1 = \frac{\gamma}{2} w_Y, \tag{17}$$

$$\dot{q}_2 = \frac{\gamma}{2} w_X, \tag{18}$$

with

$$w = \sqrt{\frac{\mu}{a}}, \tag{19}$$

$$h = \sqrt{c^2 - 2r^2 U}, \tag{20}$$

$$d = \frac{h - c}{r^2}, \tag{21}$$

$$\mathcal{E}_Z = \sum_{n \geq 2} (1 - n) U_n, \tag{22}$$

$$F_h = -\frac{\delta}{\gamma r} \frac{\partial U}{\partial \hat{z}}, \tag{23}$$

$$w_X = \frac{X}{h} F_h, \qquad w_Y = \frac{Y}{h} F_h, \qquad w_h = w_X q_1 - w_Y q_2. \tag{24}$$

This formulation is already structurally suitable for averaging because:

1. the system is autonomous;
2. the fast angle is explicit;
3. $\dot{\nu} = 0$ exactly;
4. all forcing enters through $r(K)$, $X(K)$, $Y(K)$, and $\hat{z}(K)$.

4 First-Order Orbit Averaging in K

4.1 Averaging operator

To first order in the zonal perturbation, the unperturbed fast flow is

$$\dot{K}_0 = \frac{w}{r}, \qquad \frac{dt}{dK} = \frac{r}{w}. \tag{25}$$

Hence the orbital average of any 2π-periodic quantity $f(K)$ is

$$\langle f \rangle_K = \frac{1}{T} \int_0^T f \, dt = \frac{1}{2\pi a} \int_0^{2\pi} r(K) \, f(K) \, dK. \tag{26}$$

Let the slow state be

$$\boldsymbol{z} = (\nu, p_1, p_2, q_1, q_2). \tag{27}$$

Then the exact first-order averaged slow flow of the autonomous zonal problem is

$$\dot{\bar{\nu}} = 0, \tag{28}$$

$$\dot{\bar{p}}_1 = \frac{1}{2\pi \bar{a}} \int_0^{2\pi} \bar{r} \left[\bar{p}_2 (\bar{d} - \bar{w}_h) + \frac{1}{\bar{c}} \left(\frac{\bar{X}}{\bar{a}} + 2\bar{p}_2 \right) \bar{\mathcal{E}}_Z \right] dK, \tag{29}$$

$$\dot{\bar{p}}_2 = \frac{1}{2\pi \bar{a}} \int_0^{2\pi} \bar{r} \left[\bar{p}_1 (\bar{w}_h - \bar{d}) - \frac{1}{\bar{c}} \left(\frac{\bar{Y}}{\bar{a}} + 2\bar{p}_1 \right) \bar{\mathcal{E}}_Z \right] dK, \tag{30}$$

$$\dot{\bar{q}}_1 = \frac{1}{2\pi \bar{a}} \int_0^{2\pi} \bar{r} \, \frac{\bar{\gamma}}{2} \, \bar{w}_Y \, dK, \tag{31}$$

$$\dot{\bar{q}}_2 = \frac{1}{2\pi \bar{a}} \int_0^{2\pi} \bar{r} \, \frac{\bar{\gamma}}{2} \, \bar{w}_X \, dK. \tag{32}$$

All barred intermediate quantities are evaluated from the frozen slow state $\bar{\boldsymbol{z}}$ and the fast phase K via the unperturbed relations (6)-(9) and (10).

4.2 Interpretation

Equations (29)-(32) are already a valid first-order mean-element model for any autonomous zonal expansion. They are not yet a closed analytical solution, but they are an exact one-revolution quadrature formula in the variables used by the present branch.

This is the right point of departure for research, because it separates the problem into two layers:

1. a *formal averaging layer*, which is already solved by (26)-(32);
2. a *closure layer*, where one asks whether the integrals can be reduced to explicit analytical expressions.

5 Closed First-Order J_2 Secular Model

5.1 Why J_2 is special

For the pure J_2 potential,

$$U_{J_2} = -\frac{A}{r^3}\left(1 - 3\hat{z}^2\right), \qquad A = \frac{\mu J_2 R_e^2}{2}, \tag{33}$$

the averaged disturbing function is exceptionally simple. Using the standard Keplerian averages

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{a^3\beta^3}, \qquad \left\langle \frac{\sin^2 u}{r^3} \right\rangle = \frac{1}{2a^3\beta^3}, \tag{34}$$

with $u = \omega + f$, one obtains

$$\bar{U}_{J_2} = -\frac{A}{2a^3\beta^3}\left(3\cos^2 i - 1\right) = -\frac{A}{2a^3\beta^3}\left(3\frac{\delta^2}{\gamma^2} - 1\right). \tag{35}$$

Equation (35) depends only on a, β, and i. It has no dependence on the node or pericenter angles. Consequently, at first order in J_2,

$$\dot{\bar{\nu}} = 0, \qquad \dot{\bar{g}} = 0, \qquad \dot{\bar{Q}} = 0. \tag{36}$$

Thus the magnitudes of (p_1, p_2) and (q_1, q_2) are constant and the secular dynamics reduce to pure rotations.

5.2 Secular rates

Let

$$c_i = \cos i = \frac{\delta}{\gamma}, \qquad p = a\beta^2. \tag{37}$$

Within first-order singly averaged J_2 theory, the secular drift of the mean GEqOE angular variables coincides with the classical J_2 secular drift written in the GEqOE angles Ψ and Ω. At first order in J_2, the secular nodal rate is

$$\omega_{\Omega,J_2} = -\frac{3}{2}\nu J_2\left(\frac{R_e}{p}\right)^2 c_i = -\frac{3A}{ca^3\beta^3}\frac{\delta}{\gamma}, \tag{38}$$

and the secular rate of the generalized longitude of pericenter is

$$\omega_{\Psi,J_2} = \frac{3}{4}\nu J_2\left(\frac{R_e}{p}\right)^2\left(5c_i^2 - 2c_i - 1\right) = \frac{3A}{2ca^3\beta^3}\left(5\frac{\delta^2}{\gamma^2} - 2\frac{\delta}{\gamma} - 1\right). \tag{39}$$

Because

$$p_1 = g\sin\Psi, \qquad p_2 = g\cos\Psi, \qquad q_1 = Q\sin\Omega, \qquad q_2 = Q\cos\Omega, \tag{40}$$

the first-order mean GEqOE system for J_2 becomes

$$\dot{\bar{\nu}} = 0, \tag{41}$$
$$\dot{\bar{p}}_1 = \omega_{\Psi,J_2}\,\bar{p}_2, \tag{42}$$
$$\dot{\bar{p}}_2 = -\omega_{\Psi,J_2}\,\bar{p}_1, \tag{43}$$
$$\dot{\bar{q}}_1 = \omega_{\Omega,J_2}\,\bar{q}_2, \tag{44}$$
$$\dot{\bar{q}}_2 = -\omega_{\Omega,J_2}\,\bar{q}_1. \tag{45}$$

Equations (41)-(45) are autonomous and closed. They provide exactly the kind of reduced slow-flow model that is suitable for fast metric computation.

5.3 Closed-form integration

Because the coefficients in (42)-(45) are constant along the secular J_2 flow, the system is explicitly integrable. Define the mean angular variables

$$\bar{\Psi} = \operatorname{atan2}(\bar{p}_1, \bar{p}_2), \qquad \bar{\Omega} = \operatorname{atan2}(\bar{q}_1, \bar{q}_2). \tag{46}$$

Then

$$\dot{\bar{\Psi}} = \omega_{\Psi,J_2}, \qquad \dot{\bar{\Omega}} = \omega_{\Omega,J_2}, \tag{47}$$

so that

$$\bar{\Psi}(t) = \bar{\Psi}_0 + \omega_{\Psi,J_2}(t-t_0), \qquad \bar{\Omega}(t) = \bar{\Omega}_0 + \omega_{\Omega,J_2}(t-t_0). \tag{48}$$

Let

$$\bar{g}_0 = \sqrt{\bar{p}_1(t_0)^2 + \bar{p}_2(t_0)^2}, \qquad \bar{Q}_0 = \sqrt{\bar{q}_1(t_0)^2 + \bar{q}_2(t_0)^2}. \tag{49}$$

The closed secular solution is therefore

$$\bar{\nu}(t) = \bar{\nu}_0, \tag{50}$$
$$\bar{p}_1(t) = \bar{g}_0 \sin\left[\bar{\Psi}_0 + \omega_{\Psi,J_2}(t-t_0)\right], \tag{51}$$
$$\bar{p}_2(t) = \bar{g}_0 \cos\left[\bar{\Psi}_0 + \omega_{\Psi,J_2}(t-t_0)\right], \tag{52}$$
$$\bar{q}_1(t) = \bar{Q}_0 \sin\left[\bar{\Omega}_0 + \omega_{\Omega,J_2}(t-t_0)\right], \tag{53}$$
$$\bar{q}_2(t) = \bar{Q}_0 \cos\left[\bar{\Omega}_0 + \omega_{\Omega,J_2}(t-t_0)\right]. \tag{54}$$

Equivalently, the two equinoctial pairs evolve by constant-speed planar rotations,

$$\begin{bmatrix}\bar{p}_1(t)\\ \bar{p}_2(t)\end{bmatrix} = \begin{bmatrix}\cos\theta_\Psi & \sin\theta_\Psi\\ -\sin\theta_\Psi & \cos\theta_\Psi\end{bmatrix}\begin{bmatrix}\bar{p}_1(t_0)\\ \bar{p}_2(t_0)\end{bmatrix}, \qquad \theta_\Psi = \omega_{\Psi,J_2}(t-t_0), \tag{55}$$

$$\begin{bmatrix}\bar{q}_1(t)\\ \bar{q}_2(t)\end{bmatrix} = \begin{bmatrix}\cos\theta_\Omega & \sin\theta_\Omega\\ -\sin\theta_\Omega & \cos\theta_\Omega\end{bmatrix}\begin{bmatrix}\bar{q}_1(t_0)\\ \bar{q}_2(t_0)\end{bmatrix}, \qquad \theta_\Omega = \omega_{\Omega,J_2}(t-t_0). \tag{56}$$

5.4 Mean longitude / phase

If a mean fast phase is also required, the natural choice is the mean generalized longitude

$$\bar{L} = \bar{M} + \bar{\Psi}. \tag{57}$$

Using the standard first-order J_2 secular rate of the mean anomaly,

$$\dot{\bar{M}} = \nu + \frac{3}{4}\,\nu J_2 \left(\frac{R_e}{a\beta^2}\right)^2 \beta\left(3c_i^2 - 1\right), \tag{58}$$

one obtains

$$\dot{\bar{L}} = \nu + \frac{3}{4}\nu J_2 \left(\frac{R_e}{a\beta^2}\right)^2 \left[\beta(3c_i^2 - 1) + 5c_i^2 - 2c_i - 1\right]. \tag{59}$$

For many maneuver-detection applications, however, the autonomous slow subsystem (41)-(45) is the more useful core object, while the phase can be reconstructed only when needed.

5.5 Numerical sanity check

A quick consistency check was performed against the current exact J_2 GEqOE propagator in this branch using a Molniya-like case

$$r_p = 6916 \text{ km}, \qquad e = 0.74, \qquad i = 63.4^\circ, \qquad \Omega = 30^\circ, \qquad \omega = 270^\circ, \qquad M_0 = 0^\circ. \tag{60}$$

The exact J_2 GEqOE system was propagated for 20 days and the unwrapped phases

$$\Psi(t) = \text{atan2}(p_1(t), p_2(t)), \qquad \Omega(t) = \text{atan2}(q_1(t), q_2(t)) \tag{61}$$

were fitted by least squares to straight lines. The fitted slopes agreed with (38) and (39) at the level of roughly 3×10^{-4} in relative error. This does not replace a formal derivation, but it is a strong indication that the closed secular model (54) is capturing the correct first-order drift of the exact GEqOE J_2 flow.

5.6 Secular versus osculating representation

To inspect the issue more directly, the exact J_2 GEqOE solution was sampled dense enough to compute per-orbit means over 40 revolutions. The diagnostic case shown below uses the same Molniya-like geometry as before but with an initial mean anomaly of 45°, because the earlier $M_0 = 0^\circ$ case sits close to a symmetry point where the short-period Ψ correction vanishes at t_0 and therefore does not stress the phase part of the inverse map. Two comparisons are the most informative:

1. the raw osculating components (p_1, p_2, q_1, q_2) against the closed secular model and the per-orbit means;
2. the unwrapped phases (Ψ, Ω) together with the residuals $g - g_{\text{sec}}$ and $Q - Q_{\text{sec}}$.

The important conclusion from Figures 1 and 2 is now sharper than before. The discrepancy was not evidence that the averaged logic itself was unsound. The secular drift law is sound, and the inverse-map strategy is also sound. The actual problem was an algebraic mistake in the intermediate formula used for $\dot{\Psi}$ inside the short-period map.

In other words, at this stage the analysis points to the following split:

1. the first-order secular law in J_2 is behaving as expected;
2. the missing correction is the short-period transformation between the osculating GEqOE produced by the full integrator and the mean GEqOE used by the secular closure;
3. the earlier bad behavior of the full numerical inverse map came from a wrong Ψ identity, not from a breakdown of the averaging approach.

With the corrected Ψ formula and the $M_0 = 45^\circ$ test case, the full numerical inverse map gives a uniform improvement across all variables:

$$\text{RMS}_{p_1} : 1.93 \times 10^{-4} \to 6.54 \times 10^{-6}, \tag{62}$$

$$\text{RMS}_{p_2} : 7.00 \times 10^{-4} \to 1.38 \times 10^{-5}, \tag{63}$$

$$\text{RMS}_{q_1} : 6.59 \times 10^{-4} \to 2.11 \times 10^{-5}, \tag{64}$$

$$\text{RMS}_{q_2} : 2.25 \times 10^{-4} \to 6.15 \times 10^{-6}, \tag{65}$$

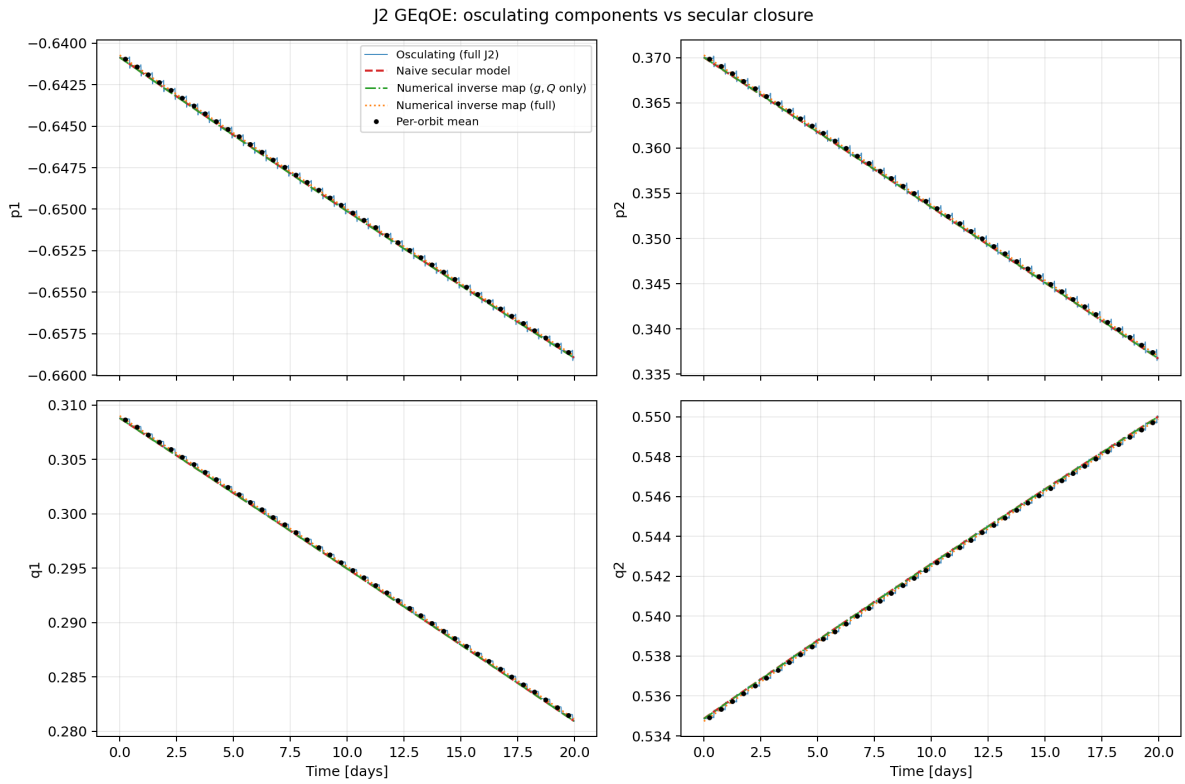

Figure 1: Exact J_2 GEqOE integration (blue), closed secular model initialized from the osculating initial condition (red dashed), the same secular model initialized with the numerically evaluated inverse map applied only to the magnitude variables (g, Q) (green dash-dot), the full numerical inverse map (orange dotted), and per-orbit means of the exact solution (black dots). In this $M_0 = 45°$ case, the full inverse map lies essentially on top of the orbit means, while the (g, Q)-only correction is not enough to remove the phase mismatch.

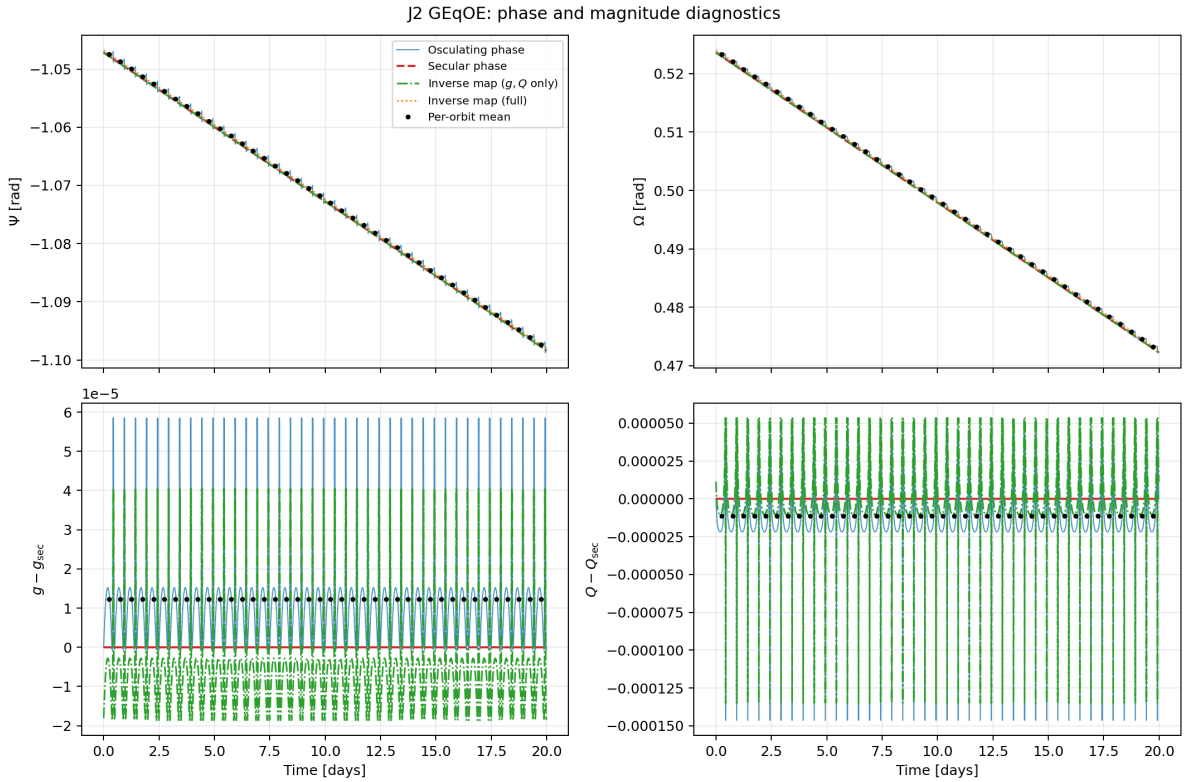

Figure 2: Phase and magnitude diagnostics for the same Molniya-like case. Top row: the unwrapped phases Ψ and Ω from the full integration are compared against the naive secular model (red dashed), the (g, Q)-only inverse map (green dash-dot), and the full inverse map (orange dotted). The full map tracks the per-orbit means closely, whereas the (g, Q)-only correction leaves the large phase bias almost unchanged. Bottom row: the residuals $g - g_{\text{sec}}$ and $Q - Q_{\text{sec}}$ show that the magnitude bias is removed by the inverse map; these panels are unchanged between the (g, Q)-only and full-map cases because the extra correction acts only on the angular variables.

while the phase mismatch drops from

$$\Psi : 7.79 \times 10^{-3} \to 8.69 \times 10^{-5}, \qquad \Omega : 7.70 \times 10^{-3} \to 2.37 \times 10^{-4}. \tag{66}$$

The initial magnitude biases move from

$$g : 1.22 \times 10^{-5} \to -5.79 \times 10^{-6}, \qquad Q : -1.14 \times 10^{-5} \to -2.17 \times 10^{-9}. \tag{67}$$

This changes the interpretation of the earlier concern. The mean-element formulation did not fail at a supposedly simple stage; the derivation of one intermediate phase identity did. Once that identity is corrected, the numerical inverse map behaves exactly as the averaged logic says it should.

5.7 What is validated and what is not

At this point, the *slow* part of the first-order formulation is on much firmer ground than it first appeared. A targeted numerical check now shows that the remaining fast-phase discrepancy was not a failure of the averaged theory, but a coordinate mismatch in the numerical implementation of the L short-period map:

1. if the full inverse map is used for (g, Q, Ψ, Ω) and the exact osculating K from the reference integration is injected, the reconstructed Cartesian position stays within sub-kilometer error over multi-orbit arcs;

2. if one instead uses the present classical mean-longitude surrogate (59) to reconstruct the fast phase without a separately validated v_1 correction for K, the Cartesian reconstruction error remains at the level of tens to hundreds of kilometers;

3. if the fast phase is rebuilt from the *exact frozen-state GEqOE* mean-longitude rate and a numerical short-period correction for L *evaluated as a function of the absolute fast angle* K, then the Cartesian reconstruction error collapses to the same sub-kilometer regime, in practice to $\mathcal{O}(10^2\,\mathrm{m})$ in the present Molniya-like test.

Thus the current bottleneck is no longer the slow secular GEqOE closure, nor the practical first-order fast-phase reconstruction itself. The real lesson is more specific: the numerical L short-period correction generated by the frozen-state GEqOE averaging is naturally a function $\ell_1(\bar{\mathbf{z}}, \bar{K})$ of the *absolute* fast phase $\bar{K}$. It should not be evaluated as if it were a function of $G = K - \Psi$. Once that distinction is respected, the frozen-state GEqOE mean-L route behaves exactly as the averaged formulation suggests it should.

What remains incomplete is not the numerical first-order closure, but its most compact analytical presentation. The next derivation step is therefore to write the corresponding v_1/ℓ_1 map directly from the exact GEqOE K/L equations in a clean first-order form, and then extend that logic to higher zonal truncations and eventually thrust.

Numerically, this already suggests the right direction. The classical $\dot{\bar{L}}$ surrogate is too weak for accurate state reconstruction, whereas the exact frozen-state GEqOE average of $\dot{L}$ together with the correct K-based evaluation of ℓ_1 produces an almost complete closure. In the current Molniya-like test case:

$$\text{mean position error} \approx 88 \text{ km} \longrightarrow 0.09 \text{ km} \tag{68}$$

when one replaces the classical $\dot{\bar{L}}$ model by the exact frozen-state GEqOE $\langle \dot{L} \rangle$ together with a numerical short-period correction for L evaluated at $\bar{K}$. This confirms that the correct closure is indeed contained inside the GEqOE L/K dynamics rather than in the classical Keplerian mean-anomaly surrogate.

6 How the Mean-to-Osculating Map Works

6.1 Near-identity transformation

A complete mean-element formulation requires more than the secular equations. It also requires a *near-identity transformation* between:

1. the osculating GEqOE of the full J_2 dynamics, and
2. the mean GEqOE propagated by the secular model.

Write the exact first-order J_2 problem in the form

$$\dot{\boldsymbol{z}} = \varepsilon \boldsymbol{f}_1(\boldsymbol{z}, K) + O(\varepsilon^2), \tag{69}$$

$$\dot{K} = n_0(\boldsymbol{z}, K) + \varepsilon g_1(\boldsymbol{z}, K) + O(\varepsilon^2), \tag{70}$$

where

$$\boldsymbol{z} = (\nu, p_1, p_2, q_1, q_2), \qquad \varepsilon \sim J_2, \qquad n_0(\boldsymbol{z}, K) = \frac{w}{r}. \tag{71}$$

The complete averaged formulation is then sought in the form

$$\boldsymbol{z}_{\text{osc}} = \bar{\boldsymbol{z}} + \varepsilon \boldsymbol{u}_1(\bar{\boldsymbol{z}}, \bar{K}) + O(\varepsilon^2), \tag{72}$$

$$K_{\text{osc}} = \bar{K} + \varepsilon v_1(\bar{\boldsymbol{z}}, \bar{K}) + O(\varepsilon^2), \tag{73}$$

with periodic short-period corrections $\boldsymbol{u}_1$ and v_1.

The usual normalization is

$$\frac{1}{2\pi}\int_0^{2\pi} \boldsymbol{u}_1(\bar{\boldsymbol{z}}, \bar{K})\, d\bar{K} = \boldsymbol{0}, \qquad \frac{1}{2\pi}\int_0^{2\pi} v_1(\bar{\boldsymbol{z}}, \bar{K})\, d\bar{K} = 0, \tag{74}$$

which fixes the decomposition into mean and periodic parts.

6.2 Homological equations

Substituting (72)-(73) into (70) and matching terms at first order gives the homological equation for the slow variables:

$$n_0(\bar{\boldsymbol{z}}, \bar{K}) \frac{\partial \boldsymbol{u}_1}{\partial \bar{K}} = \boldsymbol{f}_1(\bar{\boldsymbol{z}}, \bar{K}) - \bar{\boldsymbol{f}}_1(\bar{\boldsymbol{z}}), \tag{75}$$

where $\bar{\boldsymbol{f}}_1$ is the averaged slow drift. Therefore

$$\frac{\partial \boldsymbol{u}_1}{\partial \bar{K}} = \frac{\bar{r}}{\bar{w}} \left[\boldsymbol{f}_1(\bar{\boldsymbol{z}}, \bar{K}) - \bar{\boldsymbol{f}}_1(\bar{\boldsymbol{z}}) \right]. \tag{76}$$

Once the secular drift is known, the short-period correction of the slow variables is obtained by a direct quadrature in $\bar{K}$:

$$\boldsymbol{u}_1(\bar{\boldsymbol{z}}, \bar{K}) = \int^{\bar{K}} \frac{\bar{r}(\phi)}{\bar{w}} \left[\boldsymbol{f}_1(\bar{\boldsymbol{z}}, \phi) - \bar{\boldsymbol{f}}_1(\bar{\boldsymbol{z}}) \right] d\phi - (\text{constant chosen to enforce (74)}). \tag{77}$$

The phase correction is slightly more delicate because the unperturbed fast rate $n_0 = w/r$ depends on $\bar{K}$. The first-order phase homological equation is

$$n_0 \frac{\partial v_1}{\partial \bar{K}} - \frac{\partial n_0}{\partial \bar{K}} v_1 = g_1 - \bar{g}_1 + \frac{\partial n_0}{\partial \bar{\boldsymbol{z}}} \cdot \boldsymbol{u}_1, \tag{78}$$

which can be rewritten as

$$\frac{\partial}{\partial \bar{K}} \left(\frac{v_1}{n_0} \right) = \frac{g_1 - \bar{g}_1 + \dfrac{\partial n_0}{\partial \bar{\boldsymbol{z}}} \cdot \boldsymbol{u}_1}{n_0^2}. \tag{79}$$

Hence v_1 is also obtained by quadrature, again with the additive constant fixed by the zero-mean normalization in (74).

6.3 Why this resolves the observed bias

The secular comparison performed above used the osculating initial GEqOE state as if it were the mean initial state. That corresponds to setting

$$\bar{\boldsymbol{z}}(t_0) = \boldsymbol{z}_{\mathrm{osc}}(t_0), \qquad \bar{K}(t_0) = K_{\mathrm{osc}}(t_0), \tag{80}$$

which ignores the $O(\varepsilon)$ correction in (72)-(73). The correct inverse initialization map is

$$\bar{\boldsymbol{z}}(t_0) = \boldsymbol{z}_{\mathrm{osc}}(t_0) - \varepsilon \boldsymbol{u}_1(\boldsymbol{z}_{\mathrm{osc}}(t_0), K_{\mathrm{osc}}(t_0)) + O(\varepsilon^2), \tag{81}$$

$$\bar{K}(t_0) = K_{\mathrm{osc}}(t_0) - \varepsilon v_1(\boldsymbol{z}_{\mathrm{osc}}(t_0), K_{\mathrm{osc}}(t_0)) + O(\varepsilon^2). \tag{82}$$

This is exactly the missing ingredient revealed by Figures 1 and 2. The nearly constant offsets in g and Q are not evidence that the secular law is wrong. They are evidence that the inverse map (81)-(82) has not yet been applied.

6.4 Practical meaning for the J_2 GEqOE formulation

The complete first-order J_2 mean-element model therefore has three pieces:

1. the secular flow (41)-(54);
2. the forward reconstruction map (72)-(73);
3. the inverse initialization map (81)-(82).

For the present GEqOE problem, the constructive workflow is:

1. isolate the first-order J_2 forcing terms $\boldsymbol{f}_1(\boldsymbol{z}, K)$ and $g_1(\boldsymbol{z}, K)$ from the exact GEqOE J_2 equations;
2. subtract their averaged parts;
3. integrate the residuals with (77) and (79);
4. fix the integration constants by the zero-mean conditions (74).

After this step, the formulation is complete in the standard singly averaged sense: one can move from osculating initial conditions to mean initial conditions, propagate the mean system, and reconstruct the osculating state at any epoch to first order in J_2.

6.5 Constructive first-order J_2 map in anomaly variables

The previous discussion becomes constructive once the J_2 forcing is written in variables adapted to the unperturbed ellipse. Introduce

$$G = K - \Psi, \qquad \omega = \Psi - \Omega, \qquad s_i = \sin i = \frac{2Q}{\gamma}, \qquad c_i = \cos i = \frac{\delta}{\gamma}. \tag{83}$$

Then the unperturbed geometry is

$$r = a(1 - g\cos G), \tag{84}$$

$$\cos f = \frac{\cos G - g}{1 - g\cos G}, \tag{85}$$

$$\sin f = \frac{\beta \sin G}{1 - g\cos G}, \tag{86}$$

$$u = \omega + f, \tag{87}$$

and therefore

$$\hat{z} = s_i \sin u. \tag{88}$$

To first order in J_2 one may set

$$h = c + O(J_2), \qquad d = -\frac{U}{c} + O(J_2^2), \qquad w_h = \frac{3A}{cr^3} c_i \hat{z}^2 + O(J_2^2), \tag{89}$$

with

$$U = -\frac{A}{r^3}\left(1 - 3s_i^2 \sin^2 u\right). \tag{90}$$

Now parameterize the slow variables by $(\nu, g, \Psi, Q, \Omega)$ rather than by $(\nu, p_1, p_2, q_1, q_2)$. Using

$$p_1 = g \sin \Psi, \quad p_2 = g \cos \Psi, \quad q_1 = Q \sin \Omega, \quad q_2 = Q \cos \Omega, \tag{91}$$

and the exact GEqOE J_2 equations, the first-order osculating dynamics become

$$\dot{\nu} = 0, \tag{92}$$

$$\dot{g} = \frac{U}{c}\beta \sin G = -\frac{A\beta}{cr^3}\left(1 - 3s_i^2 \sin^2 u\right) \sin G, \tag{93}$$

$$\dot{Q} = -\frac{6AQc_i}{cr^3} \sin u \cos u, \tag{94}$$

$$\dot{\Omega} = -\frac{6Ac_i}{cr^3} \sin^2 u, \tag{95}$$

$$\dot{\Psi} = d - w_h - \frac{U}{c}\left(1 + \frac{\cos G}{g}\right) + O(J_2^2). \tag{96}$$

Equations (93)-(96) already show the main structural point: once expressed through (G, u), the first-order forcing is a trigonometric-rational function of the anomaly.

The $\dot{\Psi}$ expression must be derived directly from the exact GEqOE p_1, p_2 equations,

$$\dot{\Psi} = \frac{p_2 \dot{p}_1 - p_1 \dot{p}_2}{g^2}, \tag{97}$$

not by shortcut analogy. For the zonal GEqOE right-hand side this gives

$$\dot{\Psi} = d - w_h + \frac{\mathcal{E}_Z}{c}\left(1 + \frac{\cos G}{g}\right), \tag{98}$$

and therefore (96) for J_2 with $\mathcal{E}_Z = -U$. The earlier mismatch in the numerical inverse-map test was traced to an algebraic error in this step: the incorrect factor $\cos G + g$ was used instead of $1 + \cos G/g$.

Multiplying by

$$\frac{dt}{dG} = \frac{r}{w} + O(J_2) \tag{99}$$

gives the anomaly-form equations for the short-period map. For the two magnitude variables:

$$\frac{dg}{dG} = -\frac{A}{\mu r^2}\left(1 - 3s_i^2 \sin^2 u\right) \sin G, \tag{100}$$

$$\frac{dQ}{dG} = -\frac{6AQc_i}{\mu \beta r^2} \sin u \cos u. \tag{101}$$

For the two angles:

$$\frac{d\Omega}{dG} = -\frac{6Ac_i}{\mu \beta r^2} \sin^2 u, \tag{102}$$

$$\frac{d\Psi}{dG} = \frac{r}{w}\dot{\Psi} = \frac{r}{w}\left[d - w_h - \frac{U}{c}\left(1 + \frac{\cos G}{g}\right)\right]. \tag{103}$$

The mean-to-osculating short-period corrections are now obtained by subtracting the secular pieces and integrating over G:

$$\mathcal{G}_1'(G) = \frac{dg}{dG}, \qquad \langle \mathcal{G}_1 \rangle = 0, \tag{104}$$

$$\mathcal{Q}_1'(G) = \frac{dQ}{dG}, \qquad \langle \mathcal{Q}_1 \rangle = 0, \tag{105}$$

$$\mathcal{N}_1'(G) = \frac{d\Omega}{dG} - \frac{r}{w}\,\omega_{\Omega,J_2}, \qquad \langle \mathcal{N}_1 \rangle = 0, \tag{106}$$

$$\mathcal{P}_1'(G) = \frac{d\Psi}{dG} - \frac{r}{w}\,\omega_{\Psi,J_2}, \qquad \langle \mathcal{P}_1 \rangle = 0. \tag{107}$$

Hence the first-order map can be written as

$$g_{\rm osc} = \bar{g} + \mathcal{G}_1(\bar{G}) + O(J_2^2), \tag{108}$$

$$Q_{\rm osc} = \bar{Q} + \mathcal{Q}_1(\bar{G}) + O(J_2^2), \tag{109}$$

$$\Psi_{\rm osc} = \bar{\Psi} + \mathcal{P}_1(\bar{G}) + O(J_2^2), \tag{110}$$

$$\Omega_{\rm osc} = \bar{\Omega} + \mathcal{N}_1(\bar{G}) + O(J_2^2), \tag{111}$$

where $\bar{G} = \bar{K} - \bar{\Psi}$.

Finally, the GEqOE components follow immediately:

$$p_{1,\rm osc} = g_{\rm osc} \sin \Psi_{\rm osc}, \tag{112}$$

$$p_{2,\rm osc} = g_{\rm osc} \cos \Psi_{\rm osc}, \tag{113}$$

$$q_{1,\rm osc} = Q_{\rm osc} \sin \Omega_{\rm osc}, \tag{114}$$

$$q_{2,\rm osc} = Q_{\rm osc} \cos \Omega_{\rm osc}. \tag{115}$$

6.6 Clean first-order fast-phase map in L and K

The missing fast-phase piece can be written just as explicitly. Introduce the generalized longitude

$$L = K + p_1 \cos K - p_2 \sin K, \tag{116}$$

so that, for the unperturbed geometry,

$$\frac{\partial L}{\partial K} = 1 - p_1 \sin K - p_2 \cos K = \frac{r}{a}. \tag{117}$$

At first order in J_2 the forward longitude map is

$$L_{\rm osc} = \bar{L} + \ell_1(\bar{\boldsymbol{z}}, \bar{K}) + O(J_2^2), \tag{118}$$

with ℓ_1 normalized by

$$\frac{1}{2\pi} \int_0^{2\pi} \ell_1(\bar{\boldsymbol{z}}, \bar{K})\, d\bar{K} = 0. \tag{119}$$

For the conservative J_2 problem, the exact GEqOE L equation reduces at first order to

$$\dot{L} = \nu + d - w_h - \frac{U}{c}\left[\frac{1}{\alpha} + \alpha\left(1 - \frac{r}{a}\right)\right] + O(J_2^2), \tag{120}$$

and therefore

$$\ell_1'(\bar{K}) = \frac{\bar{r}}{\bar{w}}\left[\dot{L}(\bar{\boldsymbol{z}}, \bar{K}) - \langle \dot{L} \rangle_K(\bar{\boldsymbol{z}})\right], \qquad \langle \ell_1 \rangle_K = 0. \tag{121}$$

The important structural point is that ℓ_1 is naturally parameterized by the *absolute* fast angle $\bar{K}$, not by $\bar{G} = \bar{K} - \bar{\Psi}$. This is where the earlier numerical false alarm came from.

Now expand (116) with the first-order GEqOE map (72)-(73). Writing

$$p_{1,\mathrm{osc}} = \bar{p}_1 + u_{p_1}, \qquad p_{2,\mathrm{osc}} = \bar{p}_2 + u_{p_2}, \tag{122}$$

one gets

$$\ell_1 = \frac{\bar{r}}{a} v_1 + u_{p_1} \cos \bar{K} - u_{p_2} \sin \bar{K} + O(J_2^2). \tag{123}$$

Using the (g, Ψ) short-period map (111), the component corrections are

$$u_{p_1} = \mathcal{G}_1(\bar{G}) \sin \bar{\Psi} + \bar{g}\, \mathcal{P}_1(\bar{G}) \cos \bar{\Psi}, \tag{124}$$

$$u_{p_2} = \mathcal{G}_1(\bar{G}) \cos \bar{\Psi} - \bar{g}\, \mathcal{P}_1(\bar{G}) \sin \bar{\Psi}. \tag{125}$$

Substituting (124)-(125) into (123) gives the direct fast-phase correction

$$v_1(\bar{\boldsymbol{z}}, \bar{K}) = \frac{a}{\bar{r}} \left[\ell_1(\bar{\boldsymbol{z}}, \bar{K}) + \mathcal{G}_1(\bar{G}) \sin \bar{G} - \bar{g}\, \mathcal{P}_1(\bar{G}) \cos \bar{G} \right]. \tag{126}$$

Equations (118), (121), and (126) are the clean first-order GEqOE closure for the fast phase. They show two equivalent practical reconstructions:

1. reconstruct $L_{\mathrm{osc}} = \bar{L} + \ell_1$ and recover K_{osc} from the generalized Kepler relation;
2. reconstruct $K_{\mathrm{osc}} = \bar{K} + v_1$ directly from (126).

Numerically, both routes agree to first-order accuracy. In the present Molniya-like test case, the direct v_1 route gives mean/max fast-angle errors of about 6×10^{-7} and 1.5×10^{-6} rad, with mean/max Cartesian position reconstruction errors of about 4×10^{-2} and 1.2×10^{-1} km.

This provides a complete *formal* first-order formulation:

1. the secular evolution is explicit;
2. the inverse initialization map is explicit up to quadrature;
3. the forward reconstruction map is explicit up to quadrature.

The remaining algebraic step, if desired, is to evaluate the integrals in (107) in closed elementary form. Because (100)-(103) are rational-trigonometric functions of G, and because $u(G)$ is itself given by rational-trigonometric relations, these quadratures can in principle be reduced by the tangent-half-angle substitution. The resulting expressions are expected to be lengthy, but they are not structurally obstructed.

Numerically, the map is already useful in full. After correcting the $\dot{\Psi}$ identity as discussed above, the g, Q, Ψ, and Ω quadratures can all be evaluated directly and they reduce the naive mean-element mismatch by one to two orders of magnitude in the test cases examined here.

7 What Changes for General Zonal Expansions

7.1 Autonomy is preserved

For a general static zonal potential, the averaged GEqOE system remains autonomous. This is the main reason to attack the zonal problem before adding non-autonomous perturbations. The exact first-order mean system is already given by (29)-(32).

7.2 J_2 is not representative of all zonals

The J_2 closure is unusually simple because the averaged disturbing function depends only on a, e, and i. Higher zonals are not guaranteed to share that property:

1. odd zonals such as J_3 generally retain periapsis dependence in the singly averaged problem;
2. higher even zonals can also introduce more complicated slow-angle structure than the pure J_2 case;
3. the slow subsystem therefore need not remain a uniform rotation in (p_1, p_2).

What does survive is autonomy and axisymmetry. Hence the averaged vector field cannot depend on absolute time and cannot depend on the node angle by itself. In equinoctial variables this implies that the mean zonal dynamics can only depend on:

1. the invariants a, g, and Q;
2. the relative pericenter-orientation information carried by (p_1, p_2) with respect to the symmetry axis encoded by (q_1, q_2).

7.3 Recommended next derivation target

The next mathematically useful target is not "all zonals at once" but:

1. derive the closed first-order J_2 system rigorously in GEqOE notation, validating (41)-(45);
2. then evaluate the quadratures (29)-(32) term by term for $J_2 + J_3 + J_4$;
3. then decide whether a compact analytical closure exists or whether a semi-analytical quadrature is the more pragmatic model.

8 Path to a Fourier-in-K Thrust Extension

Consider adding a small non-conservative acceleration expressed in the orbital RTN frame,

$$\boldsymbol{P}(K) = P_r(K)\,\boldsymbol{e}_r + P_t(K)\,\boldsymbol{e}_t + P_h(K)\,\boldsymbol{e}_h, \tag{127}$$

with Fourier expansions in the propagated GEqOE fast angle K,

$$P_r(K) = a_{r0} + \sum_{m=1}^{M} \left(a_{rm}\cos mK + b_{rm}\sin mK\right), \tag{128}$$

$$P_t(K) = a_{t0} + \sum_{m=1}^{M} \left(a_{tm}\cos mK + b_{tm}\sin mK\right), \tag{129}$$

$$P_h(K) = a_{h0} + \sum_{m=1}^{M} \left(a_{hm}\cos mK + b_{hm}\sin mK\right). \tag{130}$$

Let $\boldsymbol{c}$ denote the stacked Fourier coefficients. If one linearizes the forced averaged problem about the autonomous J_2 slow flow and neglects the mixed J_2-thrust coupling at first pass, the mean dynamics take the form

$$\dot{\bar{\boldsymbol{z}}} = \boldsymbol{f}_{J_2}^{sec}(\bar{\boldsymbol{z}}) + \boldsymbol{B}(\bar{\boldsymbol{z}})\,\boldsymbol{c} + O(J_2\|\boldsymbol{c}\|, \|\boldsymbol{c}\|^2), \tag{131}$$

where $\boldsymbol{f}_{J_2}^{sec}$ is the autonomous slow flow (41)-(45).

The main structural reason this is promising is that in GEqOE:

1. r/a is affine in $\sin K$ and $\cos K$;
2. X/a and Y/a are affine in $\sin K$ and $\cos K$;
3. the averaging weight is also r/a.

Therefore the thrust-induced averaged kernels are low-order trigonometric polynomials in K. Before retaining J_2-thrust coupling, the resulting matrix $\boldsymbol{B}(\bar{\boldsymbol{z}})$ should depend only on a finite low-order subset of the Fourier coefficients. This is the GEqOE analogue of the classical thrust Fourier coefficient reduction, and it is the correct next target once the autonomous J_2 baseline is fully controlled.

9 Conclusions

For the autonomous zonal problem, the averaged GEqOE program is viable.

1. The current branch already contains the right dynamical variables and the right fast angle for averaging: the generalized eccentric longitude K.
2. For a general static zonal potential, the first-order mean GEqOE equations exist immediately as the autonomous quadratures (29)-(32).
3. For pure J_2, a compact autonomous secular closure is available: ν, g, and Q are constant, while (p_1, p_2) and (q_1, q_2) rotate with the explicit rates (38)-(39).
4. This J_2 closure is a credible baseline for fast maneuver metrics.
5. The natural next extension is a Fourier-in-K thrust perturbation on top of the autonomous J_2 mean flow, not a jump directly to fully general non-autonomous perturbations.

In short: the autonomous J_2 problem is the correct place to start, the general zonal problem is still structurally tractable, and the thrust-Fourier extension should be built on top of this averaged baseline rather than beside it.